

Math 550, S20
Quiz 4

Name: Answer

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Let $\mathbf{x}(t) = e^t \mathbf{i} + \sqrt{2}t \mathbf{j} + e^{-t} \mathbf{k}$ that defines a path (or curve) in $\mathbb{R}^3$.

(a) Find the parametric equations for the line tangent to the path at the point with $t = 0$.

$$\mathbf{x}'(t) = (e^t, \sqrt{2}, -e^{-t}) \Rightarrow \mathbf{x}'(0) = (1, \sqrt{2}, -1)$$

$$\mathbf{x}(0) = (1, 0, 1)$$

$\Rightarrow$ The parametric equation of the line tangent to $\mathbf{x}$ at $(1, 0, 1)$ is

$$\begin{cases} x(t) = 1 + t \\ y(t) = \sqrt{2}t \\ z(t) = 1 - t \end{cases} \quad \text{or} \quad \ell(t) = (1, 0, 1) + t(1, \sqrt{2}, -1)$$

(b) Find the unit tangent vector $\mathbf{T}(t)$ of the path.

$$\begin{aligned} \|\mathbf{x}'(t)\| &= \sqrt{(e^t)^2 + (\sqrt{2})^2 + (e^{-t})^2} = \sqrt{e^{2t} + 2 + e^{-2t}} \\ &= \sqrt{(e^t + e^{-t})^2} = e^t + e^{-t} \end{aligned}$$

$$\Rightarrow \mathbf{T}(t) = \frac{\mathbf{x}'(t)}{\|\mathbf{x}'(t)\|} = \frac{1}{e^t + e^{-t}} (e^t, \sqrt{2}, e^{-t})$$

(c) Find the length of the arc defined by the path with $0 \leq t \leq 1$.

$$\begin{aligned} L &= \int_0^1 \|\mathbf{x}'(t)\| dt \\ &= \int_0^1 (e^t + e^{-t}) dt \\ &= (e^t - e^{-t}) \Big|_0^1 \\ &= (e - e^{-1}) - (0) = e - \frac{1}{e} \end{aligned}$$

Problem 2. (4 points) Let $\mathbf{x}(t) = (\frac{3}{2}\sin 2t, 4t, \frac{3}{2}\cos 2t)$. Find the curvature of the path defined by $\mathbf{x}$ in $\mathbb{R}^3$.

$$\mathbf{x}'(t) = (3\cos 2t, 4, -3\sin 2t)$$

$$\begin{aligned}\Rightarrow \|\mathbf{x}'(t)\| &= \sqrt{(3\cos 2t)^2 + 4^2 + (-3\sin 2t)^2} \\ &= \sqrt{9\cos^2 2t + 16 + 9\sin^2 2t} = \sqrt{25} = 5\end{aligned}$$

$$\text{thus } \mathbf{T}(t) = \frac{\mathbf{x}'(t)}{\|\mathbf{x}'(t)\|} = \left(\frac{3}{5}\cos 2t, \frac{4}{5}, -\frac{3}{5}\sin 2t\right)$$

$$\mathbf{T}'(t) = \left(-\frac{6}{5}\sin 2t, 0, -\frac{6}{5}\cos 2t\right)$$

$$\begin{aligned}\Rightarrow \|\mathbf{T}'(t)\| &= \sqrt{\left(-\frac{6}{5}\sin 2t\right)^2 + 0^2 + \left(-\frac{6}{5}\cos 2t\right)^2} \\ &= \frac{6}{5}\end{aligned}$$

So we get

$$\kappa = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{x}'(t)\|} = \frac{\frac{6}{5}}{5} = \frac{6}{25}$$